Chapter 1 — Electronic Components & Materials

RRB JE CBT-2 · Electronics & Allied Engineering (CEN 05/2025, Annexure-VII C, Unit 1)

> **Syllabus text (Unit 1):** *Electronic Components & Materials: Conductors, Semi-conductor & Insulators; Magnetic materials; Jointing & Cleaning materials for U/G copper cable & OFC; Cells and Batteries (chargeable and non-chargeable); Relays, Switches, MCB & Connectors.*

📊 Weightage & Trend Box

Parameter	Value
Weight in Technical section	**~10 %** (5 of 50 technical Q in the actual 22-04-2025 CBT-2 Shift-2 paper; ≈ 10 of 100 on the official pattern)
Difficulty level	Easy–Moderate (definition + property-based; almost no lengthy numericals)
Best scoring strategy	Memorise property tables (resistivity, dielectric strength, battery chemistry, insulation classes) — these convert directly into marks

Questions actually observed in the 22-04-2025 CBT-2 paper from this unit:

1. **Dielectric strength** — definition (maximum electric field an insulator withstands without breakdown, kV/mm).
2. **Conductor material choice** — why **copper** is preferred (high conductivity **and** corrosion resistance).
3. **Mineral insulating materials** — their applications (used where **mechanical strength + insulation** are both needed: e.g., porcelain/ceramic line insulators, mica in heating appliances).
4. **Nickel–Iron (Edison) cell** — discharge chemistry ($NiOOH/Ni(OH)_3$ positive plate reduced to $Ni(OH)_2$; iron negative plate oxidised to $Fe(OH)_2$; KOH electrolyte concentration essentially unchanged).
5. **Resistor rating** — what decides the rating of a resistor (**permissible power dissipation** — wattage).

Trend takeaway: The board tests *definitions, material selection logic and cell chemistry* — exactly the content of the tables and boxed definitions in this chapter. Every one of the five 2025 questions is solved in the **PYQ Corner** (Section 12).

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1. Classification of Materials — Conductors, Semiconductors, Insulators

1.1 On the basis of conductivity / resistivity

Class	Resistivity ρ (Ω·m)	Conductivity σ (S/m)	Examples
Conductor	10^{-8} – 10^{-6}	10^{6} – 10^{8}	Ag, Cu, Al, Fe
Semiconductor	10^{-5} – 10^{6}	10^{-6} – 10^{5}	Ge, Si, GaAs, InP
Insulator	10^{8} – 10^{18}	negligible	Mica, glass, PVC, porcelain, quartz

1.2 Energy-band explanation (the *why*)

In a solid, discrete atomic levels broaden into **bands**. The two bands that matter:

- **Valence band (VB)** — highest band filled with electrons at 0 K.
- **Conduction band (CB)** — the next higher band; electrons here are free to conduct.
- **Forbidden energy gap (E_g)** — separation between VB top and CB bottom.

Material	Band picture	Energy gap E_g	Behaviour
Conductor	VB and CB **overlap** (no gap)	$E_g \approx 0$ eV	Enormous free-electron density (~10^{28}–10^{29} /m^3); conducts at all temperatures
Semiconductor	Small gap	**Si ≈ 1.12 eV, Ge ≈ 0.72 eV, GaAs ≈ 1.43 eV** (at 300 K)	Insulator at 0 K; thermal energy at room temperature lifts some electrons into CB → moderate conduction; conductivity **rises** with temperature (negative temperature coefficient of resistance)
Insulator	Large gap	$E_g > 3$ eV (diamond ≈ 5.5 eV, SiO_2 ≈ 9 eV)	Practically no electrons reach CB; negligible conduction

Key contrasts (exam favourites):

- Conductors have a **positive** temperature coefficient of resistance (resistance ↑ with T); semiconductors and insulators have a **negative** one.
- In semiconductors current is carried by **electrons and holes**; in metals by electrons only.
- **Intrinsic** semiconductor = pure ($n_i = p_i$); **extrinsic** = doped. Pentavalent donor (P, As, Sb) → **n-type** (electrons majority); trivalent acceptor (B, Al, Ga, In) → **p-type** (holes majority). Typical doping ≈ 1 impurity atom per 10^6–10^8 host atoms.
- Fermi level: middle of the gap (intrinsic); near CB (n-type); near VB (p-type).

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2. Conductor Materials

2.1 Desirable properties of a conductor material

High conductivity (low ρ), low temperature coefficient, good **corrosion resistance**, adequate mechanical strength/ductility, solderability, availability and low cost.

2.2 Resistivity table (memorise — direct questions)

Material	ρ at 20 °C (Ω·m ×10^{-8})	Temp. coeff. α (/°C)	Chief uses
Silver (Ag)	1.59 (lowest of all)	0.0038	Contact plating, precision contacts, fuses
Copper (Cu)	1.68–1.72	0.00393	Windings, cables, busbars, PCB tracks
Gold (Au)	2.44	0.0034	Connector plating (does not tarnish)
Aluminium (Al)	2.65–2.82	0.0039	Overhead lines (ACSR), light busbars
Tungsten (W)	5.6	0.0045	Lamp filaments (m.p. 3422 °C — highest of metals)
Iron (Fe)	9.7	0.005	Earth wires, structural
Nichrome (80Ni-20Cr)	≈ 100–112	0.0001	Heating elements (high ρ, oxidation-resistant up to ~1150 °C)
Manganin (Cu-Mn-Ni)	≈ 44–48	≈ 0.00001 (almost zero)	Precision/standard resistors, shunts
Constantan / Eureka (55Cu-45Ni)	≈ 49	≈ 0.00002	Thermocouples, precision resistors, strain gauges

Why copper is the standard electrical conductor (2025 PYQ): second-highest conductivity (only silver beats it) **plus** excellent **corrosion resistance**, high ductility/malleability (drawn into fine wires), good mechanical strength, easy soldering, and reasonable cost. Silver is better electrically but too costly and tarnishes (Ag_2S); iron corrodes and has ~6× the resistivity.

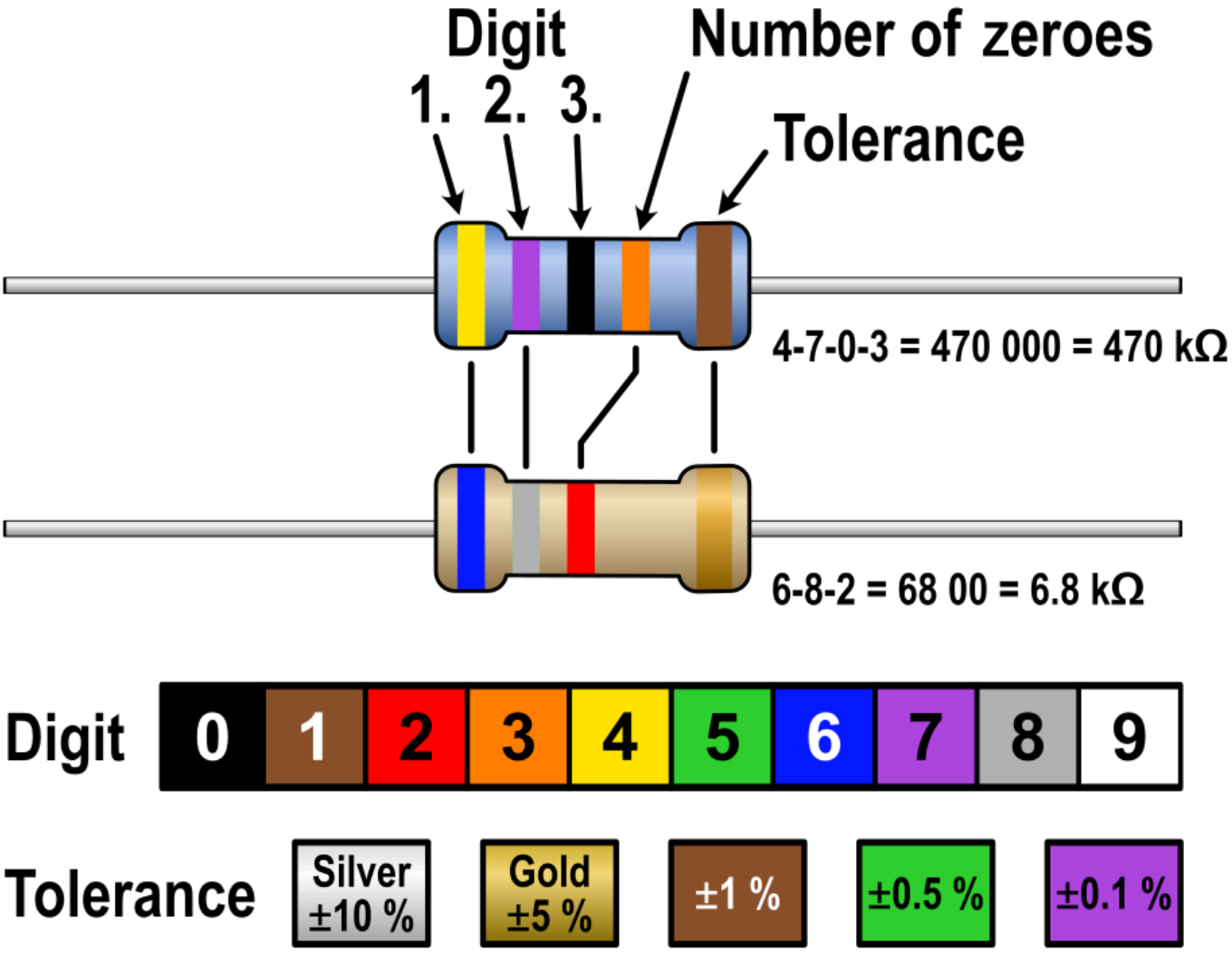
Copper vs Aluminium:

Property	Copper	Aluminium
Conductivity	100 % IACS reference	≈ 61 % of Cu
Density	8.96 g/cm^3	2.70 g/cm^3 (≈ ⅓ of Cu) — for equal conductance an Al conductor is ~50 % lighter though ~1.6× thicker
Solderability / jointing	Easy	Difficult (oxide film Al_2O_3); needs special flux/crimping
Uses	Machine windings, U/G cables, house wiring	Long-span overhead transmission (ACSR)

2.3 Solid vs stranded conductors; ACSR

- **Solid conductor:** single wire — cheapest, most rigid; used in short fixed wiring, PCB jumpers.
- **Stranded conductor:** many thin wires twisted together — **flexible**, resists fatigue from vibration/bending; standard strandings 7, 19, 37, 61 … (formula $3n^2 + 3n + 1$ layers). Used in cables, appliance cords, overhead lines.
- **ACSR (Aluminium Conductor Steel Reinforced):** central galvanised-steel strand(s) give **tensile strength** (long spans, less sag), outer aluminium strands carry the current (skin effect makes the periphery carry most AC current anyway). Standard for HV overhead transmission.
- **Bundled conductors** (2/3/4 sub-conductors per phase) at EHV reduce corona and line reactance.
- **Skin effect:** at AC, current crowds to the surface; effective resistance rises with frequency — another reason hollow/stranded/bundled conductors are used at high frequency/EHV.

3. Resistors



Resistor colour-code chart.

A resistor opposes current: $V = IR$; power dissipated $P = I^2R = V^2/R$.

3.1 Types

Type	Construction	Tolerance	Features
Carbon composition	Carbon + binder rod	±5 to ±20 %	Cheap, high noise, poor stability; withstands pulses
Carbon film	Carbon film spiralled on ceramic	±2 to ±5 %	General purpose, low cost
Metal film	Ni-Cr film on ceramic	±0.1 to ±2 %	Low noise, precise, stable — precision circuits
Metal oxide film	Tin-oxide film	±1 to ±5 %	Higher power & temperature capability
Wire-wound	Nichrome/manganin wire on ceramic former	±0.05 to ±5 %	Highest power (to hundreds of W) & precision, but **inductive** — unsuitable at high frequency (non-inductive Ayrton–Perry winding used for RF)
SMD (chip) resistor	Thick/thin film on ceramic chip (0402, 0603, 0805, 1206 sizes)	±1, ±5 %	Marked with 3/4-digit code (e.g., **472 = 47 × 10^2 = 4.7 kΩ**; 1002 = 100 × 10^2 = 10 kΩ); "0R0" = zero-ohm link
Fusible / fusistor	Designed to open safely on overload	—	Combined resistor + fuse

3.2 Colour code

B B ROY of Great Britain had a Very Good Wife →

Colour	Digit	Multiplier	Tolerance
Black	0	10^0	—
Brown	1	10^1	±1 %
Red	2	10^2	±2 %
Orange	3	10^3	—
Yellow	4	10^4	—
Green	5	10^5	±0.5 %
Blue	6	10^6	±0.25 %
Violet	7	10^7	±0.1 %
Grey	8	10^8	±0.05 %
White	9	10^9	—
Gold	—	10^{-1}	±5 %
Silver	—	10^{-2}	±10 %
No band	—	—	±20 %

- **4-band:** digit-digit-multiplier-tolerance.
- **5-band (precision):** digit-digit-digit-multiplier-tolerance.
- A 6th band, when present, gives the temperature coefficient (ppm/°C).

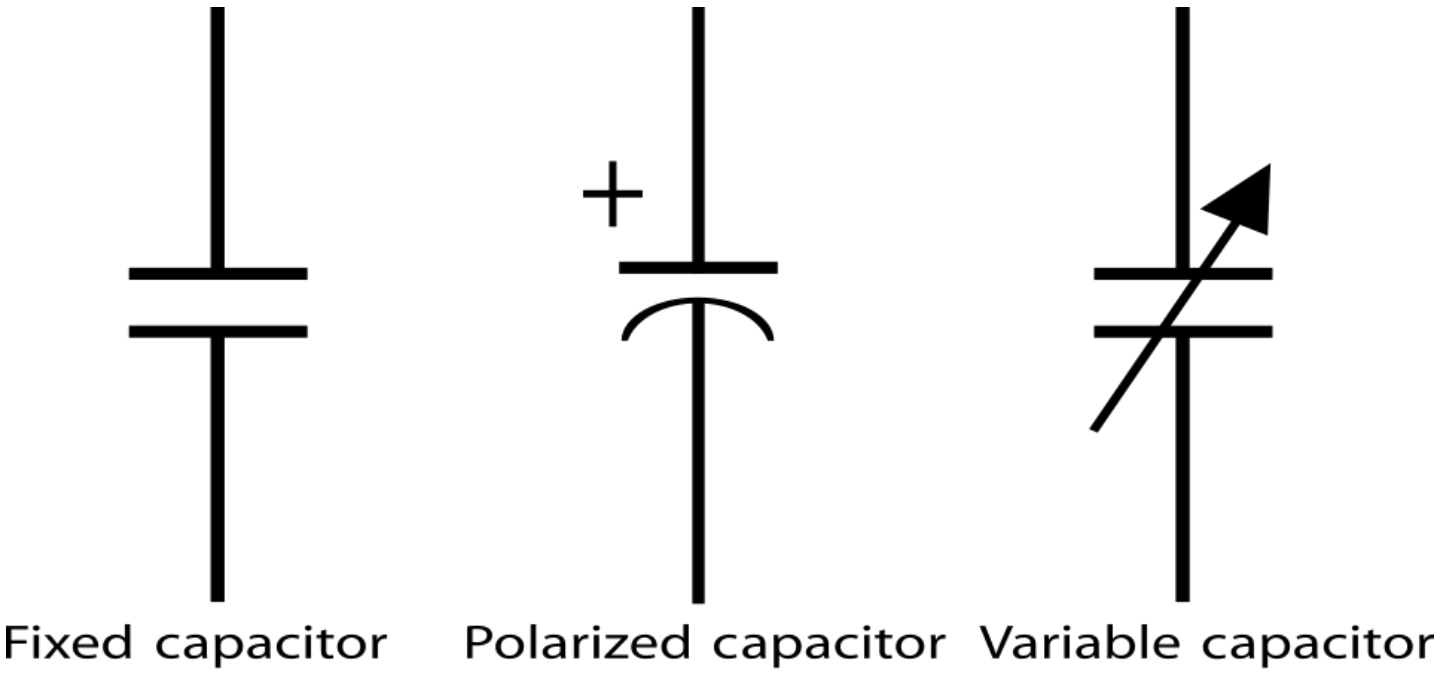
3.3 Power rating — what decides a resistor's "rating" (2025 PYQ)

> **The rating (wattage) of a resistor is determined by the maximum power it can safely dissipate as heat without its temperature exceeding the permissible limit** — i.e., by **permissible power dissipation**, which depends on physical size, surface area, materials and ambient temperature. Common ratings: ⅛ W, ¼ W, ½ W, 1 W, 2 W, 5 W, 10 W… Bigger body ⇒ higher wattage. Design rule: choose a rating ≥ 2× the expected dissipation (derating), because rated power falls at high ambient temperature.

3.4 Non-linear / special resistors

Device	Behaviour	Material	Applications
LDR (light-dependent resistor / photoresistor)	Resistance **falls** as illumination rises (MΩ dark → few hundred Ω bright)	CdS, CdSe	Street-light control, light meters, alarms
Thermistor NTC	Resistance **falls** with temperature (exponential; $R = R_0 e^{\beta(1/T - 1/T_0)}$)	Sintered Mn/Ni/Co oxides	Temperature sensing, inrush-current limiting
Thermistor PTC	Resistance **rises sharply** above a switch temperature	Doped $BaTiO_3$	Over-current/over-temperature protection, self-regulating heaters, CRT degaussing
Varistor (VDR / MOV)	Resistance falls steeply above a threshold voltage (symmetrical, $V = kI^{\beta}$)	ZnO (metal-oxide)	Surge/transient (lightning) protection across supply lines
Potentiometer / preset / rheostat	Manually variable	Carbon/cermet/wire	Volume controls, calibration, current control

4. Capacitors



Common capacitor types.

A capacitor stores charge/energy in the electric field between plates: **$C = \varepsilon A/d = \varepsilon_0\varepsilon_r A/d$**; Q = CV.

Energy stored: $W = \frac{1}{2}CV^2 = \frac{1}{2}QV = Q^2/2C$ (joules).

4.1 Two definitions you must not confuse (2025 PYQ)

> **Dielectric constant (relative permittivity, ε_r):** the ratio of the permittivity of the dielectric to that of free space ($\varepsilon/\varepsilon_0$); equivalently, the factor by which capacitance increases when the dielectric replaces vacuum. Dimensionless. (Air ≈ 1.0006, paper ≈ 2–4, mica ≈ 5–7, ceramic 6 → several thousand, water ≈ 80.)

> **Dielectric strength:** the **maximum electric field (potential gradient) an insulating material can withstand without breakdown**, i.e., without losing its insulating property. Units **kV/mm** (or V/m). It is a *breakdown* property, not a *storage* property.

Typical dielectric strengths: **air ≈ 3 kV/mm**, transformer oil ≈ 10–15 kV/mm (test ≥ 30 kV across a 2.5 mm gap), paper (impregnated) ≈ 15–25 kV/mm, porcelain ≈ 10–15 kV/mm, glass ≈ 20–30 kV/mm, **mica ≈ 50–120 kV/mm (among the highest)**, polyethylene/XLPE ≈ 20–40 kV/mm, PVC ≈ 15–30 kV/mm, ceramic ≈ 5–10 kV/mm, SF_6 gas ≈ 8–9 kV/mm at 1 atm (≈ 2.5–3× air).

4.2 Capacitor types

Type	Dielectric	Range	Polarity	Key features / uses
Ceramic (disc / MLCC)	Ceramic (Class 1 NP0/C0G stable; Class 2 X7R/Y5V high-k)	pF → µF	Non-polar	Cheap, small, RF decoupling/bypass; Class 2 values drift with V & T
Mica (silvered)	Mica sheets	pF → nF	Non-polar	Very stable, precise, low loss, high dielectric strength — RF tuned circuits, HV
Paper	Kraft paper impregnated with oil/wax	nF → µF	Non-polar	Older motor/power-factor use; superseded by film
Plastic film	Polyester (Mylar), polypropylene, polystyrene, PTFE	nF → µF	Non-polar	Low loss, self-healing (metallised film), timing, audio, snubbers, AC motor-run
Aluminium electrolytic	Al_2O_3 film formed on etched Al foil, wet electrolyte	1 µF → 10^4 µF+	**Polarised** (reverse connection destroys it)	Huge C/volume; power-supply filtering/smoothing; high leakage & tolerance (−20/+80 %); limited life (electrolyte dries)
Tantalum	Ta_2O_5 on sintered tantalum	0.1 → few hundred µF	**Polarised**	Smaller, stabler, lower leakage than Al electrolytic; low ESR; fails short if reversed/over-volted
Supercapacitor (EDLC)	Electric double layer on activated carbon	0.1 F → 1000s F	Polarised	Memory backup, energy buffering; low voltage/cell (~2.5–2.7 V)
Variable (gang / trimmer)	Air / mica / plastic	pF	Non-polar	Radio tuning, circuit alignment

4.3 Combinations

- **Parallel:** C_eq = $C_1 + C_2 + \ldots$ (capacitances add; same V).
- **Series:** 1/C_eq = $1/C_1 + 1/C_2 + \ldots$ (same Q; voltage divides inversely as C). Two in series: C_eq = $C_1C_2/(C_1+C_2)$.
- (Note this is the **opposite** of resistors.)

4.4 Practical parameters

Working voltage (choose ≥ 1.5–2× circuit voltage), tolerance, leakage current, **ESR** (equivalent series resistance — low ESR needed in SMPS filters), dissipation factor tan δ, temperature coefficient. Capacitor 3-digit code like resistor SMD: **104 = 10 × 10^4 pF = 100 nF = 0.1 µF**.

Applications recap: filtering/smoothing (electrolytic), coupling/DC-blocking, decoupling/bypass (ceramic 0.1 µF), tuning (variable/mica), timing (film, with R: τ = RC), power-factor correction, motor start/run, energy storage (photo-flash, defibrillator), snubbers.

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5. Inductors & Transformers as Components

5.1 Inductors

Store energy in a magnetic field: **$W = \frac{1}{2}LI^2$**; v = L di/dt. Unit **henry (H)**.

- **Inductance of a solenoid:** $L = \mu_0\mu_r N^2 A/l$ — increases with turns^2 and core permeability.
- **Types:** air-core (RF, no saturation), iron-core laminated (AF/power chokes), **ferrite-core** (SMPS, RF — low eddy loss), toroidal (self-shielding, high efficiency), moulded/SMD chip inductors, variable (slug-tuned).
- **Series:** $L_{eq} = L_1 + L_2$ (+2M if mutually coupled aiding, −2M opposing). **Parallel:** $1/L_{eq} = 1/L_1 + 1/L_2$ (no coupling).
- **Quality factor Q = ωL/R** — figure of merit; higher Q ⇒ sharper tuning.
- Applications: filters/chokes, tuned circuits ($f_0 = 1/2\pi\sqrt{LC}$), SMPS energy storage, RFI suppression (ferrite beads).

5.2 Transformers (component view)

Static device transferring AC power by mutual induction: **$V_2/V_1 = N_2/N_1 = I_1/I_2$** (ideal).

- EMF equation: **$E = 4.44\, f\, N\, \Phi_m$**.
- Small-signal component types: mains (50 Hz laminated **CRGO** core), audio (impedance matching: $Z_1/Z_2 = (N_1/N_2)^2$), pulse, RF/IF (ferrite/air core), isolation (1:1, safety), auto-transformer (single tapped winding — no isolation), current transformer (CT — never open-circuit the secondary while energised).
- Losses: **core/iron losses** (hysteresis + eddy — constant) and **copper loss I^2R** (load-dependent); maximum efficiency when copper loss = iron loss.

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6. Insulating Materials

6.1 Required properties

High resistivity, high **dielectric strength**, low dielectric loss (tan δ), adequate mechanical strength, thermal endurance, resistance to moisture/chemicals/ageing.

6.2 Thermal classification of insulation (IS/IEC 60085) — memorise

Class	Max. operating temp.	Typical materials
Y	90 °C	Un-impregnated paper, cotton, silk
A	105 °C	Impregnated paper/cotton, oil-immersed insulation
E	120 °C	Enamelled wire coverings, polyester film
B	130 °C	Mica, glass fibre, asbestos with organic binder
F	155 °C	Mica/glass fibre with epoxy/alkyd binders
H	180 °C	Mica/glass fibre with silicone binder; silicone rubber
C	> 180 °C	Mica, porcelain, ceramics, glass, quartz, PTFE

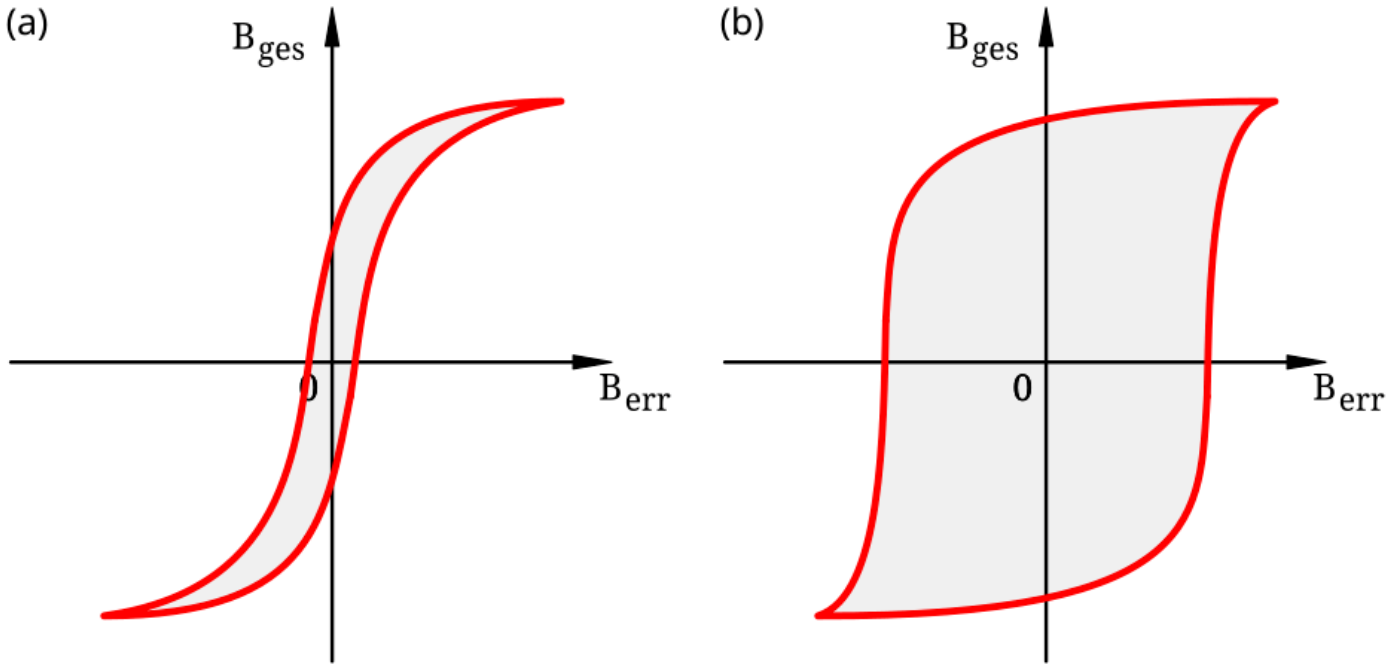
(Trick: **Y-A-E-B-F-H-C = 90-105-120-130-155-180-above 180**.)

6.3 Important insulating materials & applications

Material	Nature	Key properties	Applications
Mica (muscovite/phlogopite)	Mineral	Splits into thin flexible sheets; dielectric strength up to ~50–120 kV/mm; heat-proof to ~600 °C+; low loss	Commutator segment separation, capacitor dielectric, **heating-appliance element support (iron/toaster)**, insulation washers
Porcelain / ceramic	Mineral (fired clay + quartz + feldspar)	**High mechanical (compressive) strength + insulation together**; weather/arc resistant; brittle in tension	**Overhead-line insulators (pin, disc/suspension, strain, shackle)**, bushings, fuse bodies, resistor formers
Steatite / alumina ceramics	Mineral	High-frequency low loss; hard	RF coil formers, spark-plug insulators (alumina), substrates
Glass (toughened)	Mineral	Transparent (cracks visible), dielectric strength 20–30 kV/mm	Suspension-line insulators, lamp envelopes, glass-fibre insulation
Asbestos *(legacy)*	Mineral fibre	Heat-proof	Older heater/arc-chute insulation (now avoided — health hazard)
Rubber (natural/synthetic, silicone)	Organic/polymer	Flexible	Wire covering, gloves/mats (safety), silicone for class-H
PVC	Thermoplastic polymer	Cheap, tough, flame-retardant, moisture-proof; but max ~70–105 °C	LT cable and house-wiring insulation & sheathing
Polyethylene (PE) / XLPE	Polymer (XLPE = cross-linked)	Very low loss; XLPE withstands 90 °C continuous (250 °C short-circuit)	**HV/EHV underground cable insulation (XLPE)**, coaxial-cable dielectric (PE)
PTFE (Teflon)	Fluoropolymer	$\varepsilon_r \approx 2.1$, works −200…+260 °C, chemically inert	RF coax, aerospace wiring, non-stick bushings
Bakelite (phenolic)	Thermoset	Hard, machinable	Switch/plug bodies, terminal boards, older PCBs (FR-2); FR-4 = glass-epoxy
Transformer (mineral) oil	Liquid	Dielectric strength ≥ 30 kV per 2.5 mm gap (BDV test); also a **coolant**	Transformer/CB insulation & cooling; tested for BDV, moisture, acidity
SF_6 gas	Gas	Electronegative; dielectric strength ≈ 2.5–3× air; excellent arc quenching	Gas-insulated switchgear (GIS), HV circuit breakers *(strong greenhouse gas — handled in closed cycle)*
Air / vacuum / N_2	Gas	Air ≈ 3 kV/mm	Overhead-line spacing, vacuum interrupters (MV breakers), capacitors

Mineral insulating materials (2025 PYQ): mica, porcelain/ceramics, glass, steatite, marble, slate, asbestos, quartz. They are chosen **where the part must simultaneously insulate and bear mechanical load and/or high temperature** — overhead-line insulators and bushings (porcelain/glass), heating appliances and commutators (mica), panel boards (marble/slate legacy), high-temperature class-C insulation. Also the basis of **MICC (mineral-insulated copper-clad) cable** — copper conductors in compressed **magnesium-oxide (MgO)** powder inside a copper sheath — used for fire-survival circuits (alarms, emergency lighting).

7. Magnetic Materials



B–H hysteresis loops: soft vs hard magnetic material.

7.1 Classification by magnetic behaviour

Class	Relative permeability μ_r	Susceptibility χ	Behaviour	Examples
Diamagnetic	Slightly < 1	Small negative	Weakly **repelled** by field; magnetisation opposes H; independent of temperature	Cu, Ag, Au, Bi, Zn, water, superconductors (perfect diamagnets, $\chi = -1$)
Paramagnetic	Slightly > 1	Small positive	Weakly attracted; obeys **Curie law** $\chi \propto 1/T$	Al, Pt, Mn, Cr, O_2, Na
Ferromagnetic	$\gg 1$ ($10^2 - 10^5$)	Large positive	Strongly attracted; domains; spontaneous magnetisation; hysteresis; becomes paramagnetic above **Curie temperature** (Fe 770 °C, Ni 358 °C, Co 1131 °C)	Fe, Ni, Co, and alloys
Antiferromagnetic	≈ 1	small	Adjacent moments antiparallel, cancel	MnO, Cr
Ferrimagnetic (ferrites)	high	large	Antiparallel but unequal moments → net magnetisation; **high resistivity**	Fe_3O_4, Mn-Zn & Ni-Zn ferrites

7.2 B–H (hysteresis) loop terms

- **B** lags **H** → loop. **Retentivity/remanence (B_r):** flux density remaining when H → 0. **Coercivity (H_c):** reverse H needed to bring B to zero. **Saturation (B_sat):** further H gives no extra B.
- **Hysteresis loss per cycle ∝ loop area**; Steinmetz: **P_h = η B_max^1.6 f V** (watts).
- **Eddy-current loss:** $P_e = K_e B_{max}^2 f^2 t^2 V$ — reduced by **laminating** the core (thin varnish-insulated 0.35–0.5 mm sheets) or using high-resistivity **ferrites** at high frequency.

7.3 Soft vs hard magnetic materials

Property	Soft magnetic	Hard magnetic
Hysteresis loop	**Narrow/thin** (small area)	**Broad/fat** (large area)
Coercivity	Low (easy to demagnetise)	High (hard to demagnetise)
Retentivity usable?	Not required	High — retains magnetism
Permeability	Very high	Lower
Use	**Temporary magnets / AC cores** — transformer & motor cores, relay armatures, electromagnet cores, chokes	**Permanent magnets** — loudspeakers, PMMC meters, magnetos, small PM motors, holding magnets
Examples	Pure (soft) iron, **CRGO silicon steel**, permalloy (Ni-Fe), mumetal (shielding), amorphous metal, soft ferrites (Mn-Zn, Ni-Zn)	Carbon/tungsten/cobalt steels, **Alnico** (Al-Ni-Co-Fe), hard ferrites (barium/strontium ferrite), Sm-Co, **Nd-Fe-B (strongest)**

CRGO silicon steel: cold-rolled grain-oriented steel with ~3–4 % silicon. Silicon **raises resistivity (cuts eddy loss)** and reduces hysteresis loss; grain orientation gives highest permeability along rolling direction. Standard for **power/distribution transformer cores** (CRNGO — non-oriented — for rotating machines, where flux direction changes).

Ferrites: ceramic ferrimagnetic oxides MFe_2O_4. Very high resistivity ⇒ negligible eddy loss ⇒ ideal **above ~1 kHz to hundreds of MHz**: SMPS transformer cores, antenna rods, RFI beads, deflection yokes, memory cores (historical). Soft ferrites = Mn-Zn (to ~1 MHz), Ni-Zn (to ~100 MHz); hard ferrites = ceramic permanent magnets (cheap fridge/speaker magnets).

Alnico: cast/sintered Al-Ni-Co-Fe permanent-magnet alloy; high retentivity and best temperature stability (usable to ~500 °C); used in instruments, loudspeakers, guitar pickups.

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8. Jointing & Cleaning Materials — U/G Copper Cable & OFC

(Railway S&T context: underground jelly-filled telecom copper cables and optical-fibre cables along the track.)

8.1 Soldering materials

- **Solder** = fusible alloy that joins metals below their melting points.
- **Tin-lead (soft solder):** 60Sn-40Pb (m.p. ≈ 188 °C, electronics standard) and **63Sn-37Pb eutectic (183 °C — melts/freezes at a single temperature, no plastic/pasty range → best joints)**; 50/50 and 40/60 (plumber's, wiping joints on lead-sheathed cables).
- **Lead-free (RoHS):** SAC305 = 96.5Sn-3Ag-0.5Cu, m.p. ≈ 217 °C.
- **Hard solder / brazing:** silver solder, spelter — > 450 °C, stronger joints.
- **Flux** removes/prevents oxide so solder wets the metal:
- **Rosin (colophony)** — non-corrosive, for **electronics/telecom conductors** (RA/RMA grades).
- **Zinc chloride ("killed spirits") / acid flux** — active, for sheet-metal/galvanised work — **never on electronic joints** (corrosive residue).
- Water-soluble and no-clean fluxes in modern assembly.

- A good soldered joint is bright, concave-filleted; a **dry (cold) joint** is dull grey, high-resistance, unreliable.

8.2 Underground copper-cable jointing materials

Item	Purpose
Jelly-filled cable (PIJF) — polythene-insulated, petroleum-jelly-filled	The jelly blocks longitudinal water ingress in U/G telecom cable
Crimp/wire connectors (UY/UR/UG "B-wire"/3M Scotchlok type), jelly-filled	Splice individual copper pairs without soldering; jelly inside keeps moisture out
Cable jointing kit	Complete set: joint sleeve/closure, sealing tapes, resin or heat-shrink, connectors, abrasive strip, instructions
Heat-shrink sleeves/tubes (cross-linked polyolefin, often with hot-melt adhesive lining)	Shrink to ~½ diameter on heating (torch/hot-air) forming a sealed, insulated cover over the joint
Epoxy/polyurethane resin compound (two-part, poured into mould)	Fully encapsulates LT power-cable joints — waterproof and mechanically solid
Cold-shrink / push-on sleeves	Pre-stretched EPDM/silicone tube that contracts when the holdout is removed — no flame needed
Insulating & self-amalgamating tapes	PVC tape for covering; self-amalgamating (self-fusing) rubber tape fuses into a solid waterproof mass
Cable lugs (copper/aluminium, tinned; crimped or soldered) & **ferrules**	Terminate a conductor for bolting to equipment; bimetallic (Cu-Al) lugs stop galvanic corrosion between dissimilar metals
Cable glands (brass/nylon, with armour clamp)	Seal and anchor the cable where it enters an enclosure; earth the armour
Joint closure boxes (mechanical, re-enterable)	Protect telecom splices in manholes/pits
Denso (petrolatum) tape / bitumen compound	Corrosion protection of buried metallic sheaths

8.3 OFC (optical fibre cable) splicing & jointing

Method	How	Loss	Use
Fusion splicing	Fibre ends cleaved, aligned, welded by an **electric arc** in a fusion-splicing machine; protected by a heat-shrink **splice-protection sleeve** with steel strength member	**Lowest, ~0.01–0.1 dB**, permanent	Standard for route joints — Railways/telecom backbone
Mechanical splicing	Cleaved fibres butt-aligned in a precision V-groove/gel-filled element; **index-matching gel** fills the gap	~0.1–0.5 dB	Temporary/emergency restoration
Connectorised joints	Field/factory connectors — SC, LC, FC, ST (E2000, MPO) with PC/UPC/APC polish	0.2–0.5 dB per mated pair	Patch panels, FDMS, equipment ends

Supporting materials: **fibre cleaver** (precision 90° cleave is critical), **Kevlar/aramid-yarn scissors**, buffer/coating **stripper**, **splice closure (dome/inline)** with cable-seal heat-shrink and splice trays, loose-tube **thixotropic gel** inside the cable (water blocking), FRP/steel central strength member, HDPE outer sheath, armouring for direct burial.

8.4 Cleaning materials & agents

- **Isopropyl alcohol (IPA, ≥ 99 %)** — the standard cleaning agent for **bare fibre before splicing**, connector end-faces, and flux-residue removal on electronic joints. Fast-evaporating, residue-free.
- **Lint-free wipes / lens tissue** and **one-click connector cleaners / cleaning cassettes** for ferrule end-faces (inspect with a fibre microscope — a dirty end-face is the #1 cause of optical loss).
- **D-Gel / jelly-cleaning solvent** — removes petroleum jelly from PIJF copper pairs and cable-filling gel from loose tubes before jointing.
- **Emery/abrasive strip, wire brush, knife** — brighten copper conductors and sheath before soldering/crimping.
- **Contact cleaner sprays** (for connectors/relays), compressed air (dust), **carbon tetrachloride is obsolete/banned** (toxic, ozone-depleting) — IPA replaces it.
- Never touch or re-use a cleaned bare fibre end; always clean → cleave → splice immediately.

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9. Cells & Batteries

Cell = single electrochemical unit; **battery** = cells in series/parallel.

Primary (non-chargeable): reaction irreversible — use once. **Secondary (chargeable/accumulator):** reaction reversible by charging.

9.1 Primary cells

Cell	Anode (−)	Cathode (+)	Electrolyte	EMF	Notes
Leclanché / zinc-carbon (dry cell)	Zinc can	Carbon rod + **MnO_2 depolariser**	NH_4Cl (+ $ZnCl_2$) paste	**1.5 V**	Cheapest; polarisation (H_2 at cathode) removed by MnO_2; shelf-life short; sal-ammoniac = NH_4Cl
Alkaline (Zn-MnO_2)	Zinc powder	MnO_2	**KOH**	1.5 V	3–5× capacity of zinc-carbon, better shelf life & leak resistance
Mercury (Zn-HgO) *(phased out)*	Zn	HgO	KOH	**1.35 V — exceptionally flat/stable**	Was used as a voltage reference & in hearing aids; banned (Hg toxicity)
Silver-oxide (Zn-Ag_2O)	Zn	Ag_2O	KOH/NaOH	1.55 V	Watch button cells; very flat discharge
Zinc-air	Zn	O_2 from air	KOH	1.4 V	Highest energy density of common primaries; hearing aids
Lithium coin/cylindrical (Li-MnO_2 "CR"; Li-$SOCl_2$)	Li metal	MnO_2 (or $SOCl_2$)	Organic/non-aqueous	**3.0 V** (Li-$SOCl_2$ 3.6 V)	10-year shelf life; CMOS/RTC backup (CR2032), meters, memory retention

9.2 Secondary cells

(a) Lead-acid cell — the workhorse (Railway train lighting, telecom exchanges, substations, inverters, vehicles)

- **Construction:** positive plate **PbO_2** (chocolate brown), negative plate **spongy Pb** (grey); plates interleaved (always one more negative), separated by PVC/glass-mat separators; electrolyte **dilute H_2SO_4**; container of hard rubber/PP; vent plugs; Pb-Sb or Pb-Ca grids. Types: flooded, tubular (traction), **SMF/VRLA (sealed maintenance-free — AGM or Gel)**.
- **Overall (double-sulphation) reaction:**

$Pb + PbO_2 + 2H_2SO_4 \rightleftharpoons 2PbSO_4 + 2H_2O$ (→ discharge; ← charge)

On **discharge** both plates convert to **whitish $PbSO_4$**, acid is consumed, water is formed ⇒ **specific gravity falls**.

- **Specific gravity of electrolyte:** fully charged ≈ **1.26–1.28** (1.21–1.23 tropical/telecom float); discharged ≈ 1.13–1.18. Measured with a **hydrometer** — the best state-of-charge indicator.
- **Voltages:** nominal **2.0 V/cell**; open-circuit charged ≈ 2.1–2.2 V; **discharge cut-off 1.75–1.8 V/cell (never below)**; gassing voltage ≈ 2.4 V; float 2.15–2.25 V; equalising ≈ 2.5 V. 12 V battery = 6 cells.
- **Efficiency:** ampere-hour (quantity) efficiency ≈ 90–95 %; watt-hour (energy) efficiency ≈ 70–80 %.
- **Sulphation:** if left discharged/undercharged, $PbSO_4$ hardens into large inert crystals that will not reconvert — capacity is permanently lost. Prevention: never store discharged; recharge promptly; trickle/float charge; keep electrolyte level topped **with distilled water only** (acid is added only to correct s.g. after confirmed loss of acid).
- Other ailments: **buckling** of plates (over-discharge/heavy charge), **shedding** of active material (sediment → internal short), internal short, over-gassing (loss of water).

(b) Nickel-Cadmium (Ni-Cd)

- Positive **NiOOH**, negative **Cd**, electrolyte **KOH (≈ 21 %, s.g. ~1.2)** — *electrolyte concentration does not change* (it only transports OH^-).
- $Cd + 2NiOOH + 2H_2O \rightleftharpoons Cd(OH)_2 + 2Ni(OH)_2$. Nominal **1.2 V/cell**.
- Rugged, long life (~2000 cycles), high discharge currents, works at low temperature, flat discharge; drawbacks — **memory effect**, cadmium toxicity (restricted), lower energy density. Uses: railway coach/emergency lighting, aircraft/engine starting, power tools (legacy).

(c) Nickel-Iron (Edison) cell — ★2025 PYQ★

- **Positive electrode:** nickel oxyhydroxide **NiOOH** (often written as hydrated $Ni(OH)_3$ in older texts), in nickel-plated steel tubes with graphite/nickel flakes for conductivity.
- **Negative electrode: iron (Fe)** powder in perforated steel pockets.
- **Electrolyte: KOH ≈ 21 % (s.g. ≈ 1.2), with a little LiOH** added to improve capacity/life. As in all alkaline cells, the **electrolyte concentration/specific gravity stays practically constant** — a hydrometer cannot indicate state of charge.
- **Discharge chemistry:**
- Negative (oxidation): $Fe + 2OH^- \rightarrow Fe(OH)_2 + 2e^-$
- Positive (reduction): $2NiOOH + 2H_2O + 2e^- \rightarrow 2Ni(OH)_2 + 2OH^-$
- **Overall: $Fe + 2NiOOH + 2H_2O \rightleftharpoons Fe(OH)_2 + 2Ni(OH)_2$** (→ discharge; ← charge)
- **Voltages:** EMF charged ≈ **1.4 V**, nominal **1.2 V/cell**, discharge cut-off ≈ 1.0–1.1 V.
- **Features:** almost indestructible mechanically and electrically — tolerates overcharge, over-discharge, short-circuit and standing discharged **without sulphation-type damage**; very long life (20–30 years, thousands of cycles). Drawbacks: low energy density, heavy, poor charge retention (self-discharge 20–40 %/month), low efficiency (~65 % Wh), poor low-temperature performance, higher internal resistance and cost than lead-acid.
- Historic uses: railway train lighting & signalling, mine locomotives, standby supplies — wherever ruggedness beats efficiency.

(d) Nickel-Metal Hydride (Ni-MH)

Positive NiOOH, negative **hydrogen-absorbing alloy (MH)**, KOH electrolyte, **1.2 V/cell**; ~40 % more capacity than Ni-Cd, no cadmium, little memory effect; higher self-discharge (LSD types like Eneloop fixed this). Uses: AA/AAA rechargeables, older hybrid-car packs.

(e) Lithium-ion (Li-ion) / Li-polymer

- **Chemistry:** during **discharge**, Li^+ ions move from the graphite (LiC_6) negative through the organic electrolyte ($LiPF_6$ in carbonate solvents) and **intercalate** into the metal-oxide positive ($LiCoO_2$, NMC, $LiFePO_4$ …); electrons flow through the external circuit; charging reverses this ("rocking-chair" cell).
- Overall (LCO): $LiC_6 + CoO_2 \rightleftharpoons C_6 + LiCoO_2$.
- **Nominal 3.6–3.7 V/cell** ($LiFePO_4$ 3.2 V); charge to 4.2 V/cell with **CC-CV** method only, protection circuit (BMS) mandatory.
- **Advantages:** highest energy density (150–250 Wh/kg), light, no memory effect, low self-discharge (~2–5 %/month), long cycle life (500–2000+), high cell voltage → fewer cells.
- **Cautions:** thermal runaway if overcharged/punctured; capacity fades with age/heat. Uses: phones, laptops, EVs, modern railway applications, grid storage.

9.3 Battery terms & calculations

- **Capacity (Ah):** current × time it can deliver to cut-off at a stated rate (e.g., C_{10} = 10-hour rate, standard for lead-acid; C_{20} for automotive). Faster discharge yields **less** than rated Ah (Peukert effect).
- **C-rate:** discharge/charge current relative to capacity. For 100 Ah: 1C = 100 A, C/10 (0.1C) = 10 A, 2C = 200 A.
- **Energy (Wh) = Ah × nominal V. Energy density** = Wh/kg (gravimetric) or Wh/L (volumetric); **power density** = W/kg.
- **Cycle life:** number of charge-discharge cycles to a stated depth of discharge (DoD) before capacity falls to 80 % of rated.
- **Self-discharge, internal resistance, memory effect** (Ni-Cd), **depth of discharge (DoD)** vs state of charge (SoC).
- **Series connection:** voltages add, Ah same. **Parallel:** Ah add, voltage same.
- **Charging methods:** constant-current (Ni-Cd/Ni-MH, with −ΔV/temperature end detection), constant-voltage/**CC-CV** (lead-acid, Li-ion), taper charge, **trickle charge** (compensates self-discharge, C/50–C/100), **float charge** (battery permanently across load + charger — telecom/substation), **boost/equalising charge** (periodic controlled overcharge of flooded lead-acid to balance cells and stir electrolyte), pulse/fast charging.
- **Lead-acid maintenance drill:** keep terminals clean and petroleum-jelly-coated, electrolyte 10–15 mm above plates (top up with distilled water), check s.g. and cell voltages, keep vents clear, never leave discharged, charge in ventilated area (H_2 + O_2 evolved — explosion risk), never place metal tools across terminals.

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10. Relays

An **electromagnetic relay** is an electrically operated switch: a small coil current controls one or more contact sets switching an independent (usually larger) circuit — providing **isolation** between control and load.

10.1 Construction & operation (attracted-armature type)

Coil on an iron core + yoke; a hinged **armature** carries the moving contact against a return spring. Coil energised → magnetic pull exceeds spring force → armature attracted → contacts change over. De-energised → spring restores. Key parts: coil, core, yoke, armature, spring, contacts (silver / silver-cadmium-oxide / gold-flashed), **freewheeling diode** across a DC coil to absorb the inductive kick.

- **Pick-up (operate) value:** minimum coil current/voltage at which the relay just operates.
- **Drop-out (release/reset) value:** value at which it releases (< pick-up — hysteresis).
- **Operating time:** typically 5–20 ms (electromechanical).

10.2 Contacts, poles & throws

- **NO (normally open / make, form A):** open when de-energised, closes on operation.
- **NC (normally closed / break, form B):** closed when de-energised, opens on operation.
- **CO/changeover (form C):** one common with one NC + one NO (break-before-make).

- **Pole** = number of separate circuits switched; **throw** = positions per pole. SPST, SPDT, DPDT, 4PDT etc. — a **DPDT relay switches two independent circuits between two positions each**.

10.3 Types of relays

Type	Principle	Typical use
Attracted-armature (hinged/clapper, plunger)	Electromagnetic attraction; works on **AC or DC**; fast, instantaneous	General switching, railway signalling line/track relays, instantaneous overcurrent
Induction type (disc/cup)	Torque from two AC fluxes on an Al disc (like energy-meter); **AC only**; inherent time delay (IDMT — inverse definite minimum time)	Classic overcurrent & directional **protection** relays
Thermal relay	Bimetallic strip bends on I^2R heating	Motor **overload** protection (with contactor starters); time-lagged
Reed relay	Ferromagnetic reed contacts sealed in inert-gas glass tube, operated by coil/magnet around it	Fast (<1 ms), sealed, low-level signal switching, proximity sensing
Latching (bistable) relay	Permanent magnet/second coil holds either state with no coil power	Memory/energy-saving switching
Solid-state relay (SSR)	No moving parts — optocoupler + triac/thyristor (AC) or MOSFET (DC); zero-crossing switching versions	Silent, fast, million-cycle life, heater/PLC loads; needs heat-sink, has leakage & on-state drop
Buchholz relay	Gas-actuated float switch in transformer oil pipe	Incipient-fault protection of oil-filled transformers
Numerical/static relays	Microprocessor-based measurement	Modern substation protection

10.4 Ratings & applications

Coil rating (V, DC/AC, resistance), **contact rating (max switching V & A, AC/DC — always separately specified)**, contact resistance (mΩ), electrical/mechanical life (10^5/10^7 operations), dielectric strength coil-to-contact, operate/release time. **Applications:** control & interlocking, motor starters, **power-system protection** (overcurrent, earth-fault, differential, distance), railway signalling (track circuits, point control — where fail-safe relay logic is standard), automotive, telecom exchanges, timers/flashers.

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11. Switches

A switch makes/breaks a circuit mechanically. Described by **poles (P)** and **throws (T)** and its actuator.

Configuration	Meaning	Example use
SPST	1 circuit, on-off	Simple light switch
SPDT	1 circuit, 2 positions (common + 2 ways)	**Two-way staircase wiring** (two SPDTs), source selection
DPST	2 circuits switched together on-off	Switching line + neutral together
DPDT	2 circuits, 2 positions each	**Motor reversing**, changeover (mains/inverter)
Intermediate	Cross-over switch	Middle switch of 3-location staircase control

Actuator/mechanism types

- **Toggle switch** — lever, snap action; panel on-off.
- **Push-button** — momentary (push-to-make like a doorbell, or push-to-break) or alternate-action/latching; START (NO, green) & STOP (NC, red) in motor control.
- **Rotary switch** — one shaft, many positions/decks (e.g., multimeter range selector, fan regulator, band selector).
- **Slide switch** — small equipment on/off, mode select.
- **Rocker switch** — see-saw action; appliance mains switches.
- **Micro-switch** — snap-action switch needing tiny force/movement via lever/roller; door interlocks, limit sensing, mouse buttons; very high repeat accuracy.
- **Limit switch** — rugged micro-switch package operated by a machine part reaching a position — end-of-travel protection in cranes, lifts, machine tools.
- **DIP switch** — row of tiny slide/rocker switches in a dual-in-line package; PCB configuration/address setting.
- **Key/keypad, tactile switches** — momentary dome switches on PCBs.
- **Proximity/reed switches** — non-contact position sensing (magnetically operated reed with door magnets in alarms).
- **Selector/cam switches** — e.g., ammeter/voltmeter phase selectors, star-delta rotary starters.

Switch ratings: current & voltage (AC and DC separately — DC arcs are harder to break), contact resistance, insulation resistance, mechanical/electrical life, IP rating. Contact bounce (few ms) must be **debounced** in digital circuits (RC/flip-flop/software).

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12. MCB — Miniature Circuit Breaker

An MCB automatically disconnects a circuit on **overload** and **short-circuit**, and doubles as a manual switch. Standard: IS/IEC 60898.

12.1 Working — two tripping mechanisms in one

1. **Thermal tripping (overload):** load current heats a **bimetallic strip**; on sustained overload it bends enough to release the trip latch. Inverse-time characteristic — bigger overload trips faster (seconds→minutes).
2. **Magnetic tripping (short-circuit):** a **solenoid/electromagnet** in series; a heavy fault current instantly (< 10 ms) attracts the plunger and trips the latch.
3. The **arc** drawn at the contacts is driven by magnetic action into **arc-chute splitter plates**, split, cooled and extinguished. A trip-free mechanism ensures it trips even if the toggle is held ON.

12.2 Trip-curve classes (magnetic-trip threshold, × rated current I_n)

Curve	Instantaneous trip at	Application
B	3–5 × I_n	Resistive/lighting circuits, domestic sockets — low inrush
C	5–10 × I_n	Mixed/mildly inductive loads — motors, fluorescent banks, ACs; the general-purpose choice
D	10–20 × I_n	High-inrush loads — transformers, welding sets, large motors, X-ray
(K ≈ 8–12×, Z ≈ 2–3× — special: motor and semiconductor/instrument protection respectively)		

12.3 Ratings

- **Rated current I_n:** 0.5–125 A (common 6, 10, 16, 20, 25, 32, 40, 63 A) at 30 °C reference.
- **Poles:** SP, SP+N, DP, TP, TP+N, 4P.

- **Rated voltage** 240/415 V AC; **breaking (short-circuit) capacity:** 3, 6, 10 kA (marked in a rectangle, e.g., "10000"); **energy-limiting class 3** typical.

12.4 MCB vs rewireable fuse

Point	MCB	Fuse
Action	Trips — **reusable**, just switch back ON	Element melts — must be **replaced/rewired**
Restoration time	Seconds	Minutes + skill; risk of wrong wire size
Characteristic	Precise, factory-set thermal-magnetic	Deteriorates with age; rewireable fuse imprecise
Phase indication & safety	Clear ON/OFF/trip, no live parts touched	Live parts exposed while rewiring
Cost	Higher initial	Cheap
Speed on huge faults	~ms	HRC fuse can be even faster (current-limiting, cuts off before first peak) — hence fuses still back up breakers

12.5 Related breakers — one-liners

- **MCCB (moulded-case circuit breaker):** bigger brother — 16 A to 1600 A, breaking capacity 10–100 kA, often adjustable thermal/magnetic or electronic trip; industrial feeders.
- **ELCB (earth-leakage CB, voltage-operated):** old type tripping on voltage rise of earthed metalwork; needs sound earth connection; obsolete.
- **RCCB/RCD (residual-current CB, current-operated):** core-balance CT compares phase and neutral currents; trips when the **residual (leakage) difference exceeds 30 mA (shock protection) / 100–300 mA (fire protection)** within ~30 ms — protects against electrocution; does **not** protect against overload (combine with MCB).
- **RCBO:** RCCB + MCB overload/short-circuit protection in one module.
- **Isolator:** off-load disconnector for maintenance isolation — never open on load; a **switch-disconnector-fuse (SDF)** adds fuses.

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13. Connectors

A connector provides a **separable** electrical joint between cables, boards and equipment. Merits are judged by contact resistance, insertion/withdrawal force, mating cycles, current/voltage rating and environmental sealing.

13.1 Families

Family	Examples	Notes / uses
Audio	6.35 mm (¼") phone jack, **3.5 mm TRS/TRRS** (tip-ring-sleeve: L, R, ground, +mic), RCA/phono, **XLR (3-pin, balanced, locking)**, banana plugs	Headphones, line audio, professional mics (XLR), test leads (banana, 4 mm)
RF / coaxial	**BNC** — bayonet lock, 50/75 Ω, to ~4 GHz: oscilloscopes, video, LAN (legacy 10BASE2); **SMA** — threaded, 50 Ω, precision to ~18 GHz: antennas, RF modules; TNC (threaded BNC); **N-type** (to 11 GHz, weatherproof, base stations); UHF (PL-259/SO-239, HF radio); F-type (TV/satellite, 75 Ω); MCX/MMCX/U.FL (miniature, GPS/Wi-Fi boards)	Maintain characteristic impedance (50 Ω instruments/comms, 75 Ω video/CATV) to avoid reflections
Card-edge connectors	PCB edge fingers into a slot — PCI/PCIe, DIMM/RAM sockets	Gold-plated fingers ("goldfinger"); polarising key slot
D-subminiature (DB/DE)	**DB-9 (DE-9): RS-232 serial**; DB-25: parallel/serial legacy; DB-15/HD-15 = **VGA** (3-row)	Trapezoidal D-shell polarises the mating; screw locks
RJ family (modular)	**RJ45 (8P8C): Ethernet LAN** — T568A/T568B colour codes, ≤ 100 m Cat5e/6; RJ11/RJ12 (6P2C/6P4C): telephone	Crimped with modular crimping tool; keystone jacks
USB	Type-A, Type-B, Mini, Micro, **Type-C (reversible, 24-pin)** — USB 2.0 (480 Mb/s, 4-pin) → 3.x (extra pairs) → USB4; also powers loads (5 V; USB-PD up to 100–240 W over Type-C)	Universal peripheral/data/charging connector
Terminal blocks	Screw-clamp (barrier/PCB), spring-cage, **DIN-rail** terminals, pluggable (Phoenix) types	Panel wiring & field terminations; ferrules on stranded wire
Power & board-level	IEC 60320 C13/C14 mains inlet, Anderson, XT60, header pins/Berg, FRC/IDC ribbon (insulation-displacement), JST, ZIF/FFC for flex cables	IDC = insulation-displacement contact — no stripping needed
Circular/military & industrial	M8/M12 sensor connectors, MIL-C-5015/38999	Vibration-proof threading, IP67 sealing
Optical-fibre connectors	SC (push-pull, square), **LC (small form-factor — most common now)**, ST (bayonet), FC (threaded), MPO (multi-fibre); polish grades PC/UPC (blue) & **APC (green, 8° angle — lowest back-reflection)**	Patch panels, FTTH, transceivers (SFP uses LC)

13.2 Contact materials & plating

- **Base contact metal:** brass, phosphor bronze or beryllium copper (best spring properties).
- **Plating: gold** (flash over nickel) — does not oxidise, lowest and most stable contact resistance — for low-level signal contacts and edge fingers; **silver** — best conductivity, used in power contacts (tarnish film breaks down easily at power levels); **tin** — cheap, general power/consumer contacts (avoid mating tin with gold — fretting corrosion); **nickel** — hard barrier/underplate; palladium-nickel as gold substitute.

- Design points: wiping action self-cleans contacts on insertion; polarisation/keying prevents wrong mating; strain relief and locking (bayonet/screw/latch) preserve the joint; rated mating cycles (tin ~100s, gold ~1000s).

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14. Worked Examples (8)

Example 1 — 4-band colour code.

Bands: **Red-Violet-Orange-Gold**.

Red = 2, Violet = 7, Orange = $\times 10^3$, Gold = ±5 %.

R = 27 × 10^3 = **27 kΩ ± 5 %** (i.e., 25.65–28.35 kΩ).

Example 2 — 5-band colour code.

Bands: **Brown-Black-Black-Red-Brown**.

Digits 1, 0, 0; multiplier $\times 10^2$; tolerance ±1 %.

R = 100 × 10^2 = **10 kΩ ± 1 %**.

Example 3 — Reverse colour code.

What are the bands of a 470 Ω ± 10 % resistor (4-band)?

470 = 4, 7 × 10^1 → **Yellow-Violet-Brown-Silver**.

Example 4 — Resistor power rating (2025-PYQ style).

A 100 Ω resistor must carry 100 mA continuously. What minimum standard wattage should be chosen?

P = I^2R = $(0.1)^2$ × 100 = **1 W**. With the ×2 derating rule, select a **2 W** resistor. (Rating is set by permissible power dissipation, not by resistance value.)

Example 5 — Capacitor energy.

Energy stored in a 470 µF capacitor charged to 25 V:

W = ½CV^2 = ½ × 470×10^{-6} × 25^2 = ½ × 470×10^{-6} × 625 = **0.147 J** (≈ 147 mJ).

If voltage doubles to 50 V, energy becomes 4× = 0.587 J ($W \propto V^2$).

Example 6 — Dielectric strength.

A 0.2 mm mica sheet is to insulate 6 kV. Is it adequate if mica's dielectric strength is 50 kV/mm?

Applied stress = 6 kV / 0.2 mm = **30 kV/mm < 50 kV/mm** → withstands, with safety factor 50/30 ≈ 1.67. (For a design safety factor of 2.5, thickness needed = 6/(50/2.5) = 0.3 mm.)

Example 7 — Battery backup time.

A 12 V, 150 Ah (C_{10}) battery feeds an inverter delivering 400 W at 90 % efficiency. Estimate backup (to 100 % DoD, ideal).

Battery output power = 400/0.9 = 444.4 W → current = 444.4/12 = **37 A**.

Ideal time = 150 Ah / 37 A ≈ **4.05 h** (~4 hours). *Practically less — Peukert effect at above-C_{10} current and usable DoD limit (~80 % ⇒ ≈ 3.2 h).*

Example 8 — Cells in series-parallel.

How are 2 V, 100 Ah lead-acid cells arranged to build a 48 V, 300 Ah battery bank?

Series for voltage: 48/2 = **24 cells per string**. Parallel for capacity: 300/100 = **3 strings**. Total = 24 × 3 = **72 cells**. Energy = 48 × 300 = 14.4 kWh.

Bonus check — capacitors in series. Two capacitors 6 μF and 3 μF in series across 90 V: C_eq = (6×3)/(6+3) = 2 μF; Q = 2 μF × 90 = 180 μC on each; V_6 = 180/6 = 30 V, V_3 = 180/3 = 60 V (larger voltage across the smaller capacitor).

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15. PYQ Corner — Actual CBT-2 (22-04-2025) Questions Solved

PYQ 1. Dielectric strength of an insulating material is defined as —

(a) the ratio of its permittivity to that of free space

(b) **the maximum electric field (voltage per unit thickness) it can withstand without electrical breakdown**

(c) the resistance offered per unit volume

(d) the energy stored per unit volume

Answer: (b). Dielectric strength = maximum potential gradient the insulator sustains without losing its insulating property (breakdown). Measured in **kV/mm**. Option (a) defines *dielectric constant* — the classic distractor. Representative values: air 3 kV/mm, transformer oil ~10–15, porcelain ~10–15, mica 50+ kV/mm.

PYQ 2. Copper is widely preferred as an electrical conductor material because it has —

(a) the lowest resistivity of all metals

(b) **high electrical conductivity along with good corrosion resistance**

(c) very high resistivity and high melting point

(d) magnetic properties useful in machines

Answer: (b). Silver actually has the lowest resistivity (1.59×10^{-8} Ω·m vs copper's ~1.7×10^{-8}), so (a) is false. Copper wins on the *combination*: near-best conductivity, excellent corrosion resistance (protective oxide, no destructive tarnish problem in service), ductility for wire-drawing, strength, solderability and economy.

PYQ 3. Mineral insulating materials (mica, porcelain, glass, etc.) are chiefly used where —

(a) only flexibility is required

(b) **both good mechanical strength/heat withstand and electrical insulation are required**

(c) the operating temperature is below 90 °C only

(d) a conducting path must be provided

Answer: (b). Minerals are class-B/C insulation: porcelain/toughened glass carry the full mechanical load of overhead conductors while insulating them (pin/disc insulators, bushings); mica supports red-hot heating elements in irons and toasters and separates commutator bars; MgO fills fire-survival MICC cables. Organic insulators (class Y, ~90 °C) cannot survive these duties.

PYQ 4. In a Nickel-Iron (Edison) cell, during discharge —

(a) the positive plate PbO_2 changes to $PbSO_4$

(b) the KOH electrolyte is consumed and its specific gravity falls sharply

(c) **the nickel oxyhydroxide positive plate is reduced to $Ni(OH)_2$ while the iron negative plate is oxidised to $Fe(OH)_2$; the KOH concentration remains practically unchanged**

(d) lithium ions intercalate into a cobalt-oxide cathode

Answer: (c). Discharge reactions: $Fe + 2OH^- \rightarrow Fe(OH)_2 + 2e^-$ (negative); $2NiOOH + 2H_2O + 2e^- \rightarrow 2Ni(OH)_2 + 2OH^-$ (positive); overall **$Fe + 2NiOOH + 2H_2O \rightarrow Fe(OH)_2 + 2Ni(OH)_2$**. The KOH merely ferries OH^- ions — its specific gravity (~1.2) stays constant, so a hydrometer cannot show the state of charge (unlike lead-acid, option (b)/(a) territory). Nominal 1.2 V/cell; famously rugged — survives overcharge, deep discharge and idle-discharged storage.

PYQ 5. The rating (wattage) of a resistor is decided by —

(a) its resistance value

(b) its tolerance

(c) **the maximum power it can safely dissipate as heat (permissible power dissipation)**

(d) its temperature coefficient

Answer: (c). Two resistors of identical resistance can have very different ratings (¼ W chip vs 10 W wire-wound): the rating depends on how much heat ($P = I^2R = V^2/R$) the body can shed without exceeding its temperature limit — governed by physical size, surface area, construction material and ambient temperature, not by the ohmic value itself.

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16. Practice MCQs (25)

Q1. The forbidden energy gap of silicon at room temperature is approximately —

(a) 0.3 eV (b) 0.72 eV (c) 1.12 eV (d) 5.5 eV

Q2. The material having the lowest electrical resistivity is —

(a) copper (b) silver (c) aluminium (d) gold

Q3. Manganin is used for making standard resistances mainly because of its —

(a) high resistivity only (b) low cost (c) **very low temperature coefficient of resistance** (d) high melting point

Q4. In an ACSR conductor, the steel core primarily provides —

(a) higher conductivity (b) mechanical/tensile strength (c) corona reduction (d) lower skin effect

Q5. A resistor has bands Green-Blue-Red-Gold. Its value is —

(a) 56 kΩ ± 5 % (b) 5.6 kΩ ± 5 % (c) 560 Ω ± 5 % (d) 6.5 kΩ ± 10 %

Q6. An NTC thermistor's resistance —

(a) rises with temperature (b) falls with temperature (c) is independent of temperature (d) falls with light intensity

Q7. A MOV (metal-oxide varistor) connected across a supply protects equipment against —

(a) overload current (b) earth leakage (c) voltage surges/transients (d) low voltage

Q8. Which capacitor is polarised and must never be connected in reverse?

(a) mica (b) ceramic disc (c) aluminium electrolytic (d) polypropylene film

Q9. Two capacitors of 4 μF and 12 μF are connected in series. The equivalent capacitance is —

(a) 16 μF (b) 8 μF (c) 3 μF (d) 48 μF

Q10. The energy stored in a 100 μF capacitor charged to 200 V is —

(a) 0.2 J (b) 2 J (c) 4 J (d) 0.02 J

Q11. Class F insulation permits a maximum operating temperature of —

(a) 105 °C (b) 130 °C (c) 155 °C (d) 180 °C

Q12. XLPE is widely used as —

(a) transformer-core material (b) HV underground-cable insulation (c) fuse element (d) contact plating

Q13. A material weakly repelled by a magnetic field (μ_r slightly less than 1) is —

(a) ferromagnetic (b) paramagnetic (c) diamagnetic (d) ferrimagnetic

Q14. Transformer cores are made of CRGO silicon steel because silicon —

(a) raises permeability by lowering resistivity (b) raises resistivity, reducing eddy-current loss (c) makes the steel a permanent magnet (d) increases hysteresis loss

Q15. A permanent magnet should be made of a material having —

(a) low retentivity, low coercivity (b) high retentivity, high coercivity (c) high permeability, narrow B-H loop (d) low retentivity, high permeability

Q16. Ferrite cores are preferred for SMPS/high-frequency transformers because they have —

(a) very high electrical resistivity, hence negligible eddy loss (b) the highest saturation flux density (c) zero hysteresis (d) high mechanical strength

Q17. Eutectic tin-lead solder (63Sn-37Pb) is preferred in electronics because it —

(a) melts at the highest temperature (b) melts/solidifies at a single temperature with no pasty range (c) needs no flux (d) is mechanically strongest of all alloys

Q18. The flux recommended for soldering electronic/telecom copper joints is —

(a) zinc chloride (killed spirits) (b) hydrochloric acid (c) rosin (d) caustic soda

Q19. The splicing method giving the lowest loss (~0.01–0.05 dB) for optical fibre is —

(a) mechanical splice (b) fusion splice (c) connectorised joint (d) twisted joint

Q20. The standard cleaning agent for fibre ends and connector faces before splicing is —

(a) kerosene (b) carbon tetrachloride (c) isopropyl alcohol (d) soap water

Q21. In a fully charged lead-acid cell, the specific gravity of the electrolyte is about —

(a) 1.00 (b) 1.13 (c) 1.28 (d) 1.80

Q22. "Sulphation" in a lead-acid battery is caused by —

(a) overcharging at high rate (b) adding excess distilled water (c) leaving the battery in a discharged state for long periods (d) too high electrolyte level

Q23. The nominal voltage per cell of Ni-Cd, Ni-Fe and Ni-MH cells is —

(a) 2.0 V (b) 1.5 V (c) 1.2 V (d) 3.6 V

Q24. A C-type MCB provides instantaneous (magnetic) tripping at —

(a) 1–2 × I_n (b) 3–5 × I_n (c) 5–10 × I_n (d) 10–20 × I_n

Q25. The connector normally used for terminating an Ethernet LAN cable is —

(a) BNC (b) DB-9 (c) RJ45 (d) SMA

Answer key with brief explanations

Q	Ans	Why
1	c	Si gap ≈ 1.12 eV (Ge 0.72 eV, diamond 5.5 eV).
2	b	Silver 1.59×10^{-8} Ω·m — lowest of all metals; Cu next.
3	c	Manganin's α ≈ 10^{-5}/°C — resistance almost constant with temperature (also low thermo-EMF with Cu).
4	b	Steel = strength (long spans, low sag); outer Al strands carry the current.
5	b	5,6 × 10^2 = 5600 Ω = 5.6 kΩ; gold = ±5 %.
6	b	NTC: negative temperature coefficient — R falls exponentially as T rises.
7	c	Varistor resistance collapses above threshold voltage, clamping surges/spikes.
8	c	Electrolytics (Al, Ta) are polarised — reverse polarity destroys the oxide film.
9	c	Series: (4×12)/(4+12) = 3 µF — less than the smallest.
10	b	$W = ½CV^2 = ½\times100\times10^{-6}\times(200)^2 = ½\times10^{-4}\times4\times10^4$ = 2 J.
11	c	Y-A-E-B-F-H-C = 90-105-120-130-**155**-180->180 °C.
12	b	Cross-linked polyethylene: 90 °C continuous, low loss — standard HV U/G cable insulation.
13	c	Diamagnetics (Cu, Bi, water) are feebly repelled; χ small negative.
14	b	~3–4 % Si multiplies resistivity → eddy loss ↓ (also hysteresis ↓); grain orientation boosts µ.
15	b	Permanent magnet = hard material: broad loop, high B_r and H_c (Alnico, ferrite, Nd-Fe-B).
16	a	Ferrites are ceramic semicon/insulators — eddy currents negligible even at MHz.
17	b	Eutectic point 183 °C: no plastic range ⇒ no disturbed/dry joints while freezing.
18	c	Rosin is non-corrosive; acid/$ZnCl_2$ fluxes corrode fine conductors.
19	b	Arc-fusion welds the glass — lowest, permanent loss; mechanical ~0.1–0.5 dB.
20	c	≥99 % IPA with lint-free wipes — residue-free; CCl_4 banned/toxic.
21	c	Charged 1.26–1.28 (falls to ~1.15 discharged); read by hydrometer.

Q	Ans	Why
22	c	Standing discharged lets $PbSO_4$ harden into large non-convertible crystals — permanent capacity loss.
23	c	All three alkaline nickel systems: 1.2 V nominal (lead-acid 2.0 V, Li-ion 3.6–3.7 V).
24	c	B: 3–5×, **C: 5–10×**, D: 10–20× I_n.
25	c	RJ45 (8P8C modular) on Cat5e/6 twisted pair; BNC was for legacy coax LAN.

17. Rapid-Revision Fact Sheet

A. Energy gaps & resistivities

Item	Value
E_9: Ge / Si / GaAs	0.72 / 1.12 / 1.43 eV
E_9 insulator (diamond)	> 3 eV (≈ 5.5 eV)
ρ: Ag / Cu / Au / Al / W	1.59 / 1.7 / 2.44 / 2.7 / 5.6 ($\times 10^{-8}$ Ω·m)
ρ: Manganin / Constantan / Nichrome	≈ 45 / 49 / 100–112 ($\times 10^{-8}$ Ω·m)
Lowest ρ metal / conductor % IACS of Al	Silver / ≈ 61 %
α of manganin	~10^{-5} /°C (near zero)

B. Colour code & component formulas

Item	Value
Colour order 0→9	Black Brown Red Orange Yellow Green Blue Violet Grey White
Tolerance	Gold ±5 %, Silver ±10 %, none ±20 %, Brown ±1 %, Red ±2 %
Resistor rating decided by	**Permissible power dissipation (wattage)**
Capacitor energy / inductor energy	$W = \frac{1}{2}CV^2$ / $W = \frac{1}{2}LI^2$
Caps: series / parallel	$1/C_{eq} = \Sigma 1/C$ / $C_{eq} = \Sigma C$
SMD "472", cap "104"	4.7 kΩ; 100 nF

C. Dielectric strengths (kV/mm) & insulation classes

Material	kV/mm
Air	3
Transformer oil	10–15 (BDV ≥ 30 kV/2.5 mm)
Porcelain	10–15
Glass	20–30
XLPE / PE	20–40
Mica	**50–120 (highest common)**
SF_6 (1 atm)	≈ 8–9 (2.5–3 × air)

Class	Y	A	E	B	F	H	C
°C max	90	105	120	130	155	180	>180

D. Magnetic quick facts

Item	Fact
Soft magnetic	Narrow B-H loop, low Hc — CRGO, permalloy, soft ferrite → transformer/machine cores
Hard magnetic	Broad loop, high B_r & Hc — Alnico, hard ferrite, Nd-Fe-B → permanent magnets
Hysteresis loss	∝ f · B_max^1.6 (Steinmetz), ∝ loop area
Eddy loss	∝ f^2 B^2 t^2 — cut by lamination / ferrites
Curie temp Fe / Ni	770 °C / 358 °C
Diamagnetic examples	Cu, Ag, Bi, water (μ_r < 1)

E. Battery chemistries & voltages

Cell	Type	Nominal V	Electrolyte	Key point
Zinc-carbon (Leclanché)	Primary	1.5	NH_4Cl paste	MnO_2 = depolariser
Alkaline Zn-MnO_2	Primary	1.5	KOH	3–5× capacity of Zn-C
Mercury / Silver-oxide	Primary	1.35 / 1.55	KOH	Flat voltage (reference/watches)
Lithium CR coin	Primary	3.0	Organic	10-yr shelf life, RTC backup
Lead-acid	Secondary	**2.0**	dil. H_2SO_4, s.g. 1.26–1.28 charged	s.g. falls on discharge; sulphation if left discharged; cut-off 1.75 V
Ni-Cd	Secondary	1.2	KOH (const. s.g.)	Rugged, memory effect, Cd toxic
Ni-Fe (Edison)	Secondary	1.2 (EMF 1.4)	KOH + LiOH (const. s.g.)	$Fe + 2NiOOH + 2H_2O \rightleftharpoons Fe(OH)_2 + 2Ni(OH)_2$; 20–30 yr life; abuse-proof; high self-discharge
Ni-MH	Secondary	1.2	KOH	MH negative; no Cd; AA rechargeables
Li-ion	Secondary	3.6–3.7 (charge 4.2)	$LiPF_6$ organic	Li^+ intercalation; highest energy density; CC-CV charge; BMS needed
C-rate for 100 Ah	—	—	—	1C = 100 A; C/10 = 10 A

F. Relays, switches, MCB, connectors

Item	Fact
Relay contacts	NO (form A) / NC (form B) / changeover (form C); pick-up > drop-out
Relay types	Attracted-armature (AC/DC, instantaneous), induction (AC only, IDMT protection), thermal (motor overload), reed (sealed, fast), SSR (no moving parts), Buchholz (transformer gas)
DC relay coil must have	Freewheeling diode across it
Switch configs	SPST on-off; SPDT two-way staircase; DPDT reversing/changeover; intermediate = 3rd location
Micro/limit switch	Snap action, position sensing / end-of-travel
MCB tripping	Thermal (bimetal, overload, inverse-time) + magnetic (solenoid, short-circuit, instantaneous)
MCB curves	**B: 3–5×, C: 5–10×, D: 10–20× I_n**
MCB breaking capacity	3/6/10 kA (IS/IEC 60898)
RCCB sensitivity	30 mA (shock) / 300 mA (fire); no overload protection
RCBO	RCCB + MCB combined
BNC / SMA	Bayonet coax to ~4 GHz / threaded 50 Ω to ~18 GHz
DB-9 / HD-15 / RJ45	RS-232 serial / VGA / Ethernet 8P8C
Best contact plating (signal)	Gold (no oxide, stable low resistance); silver for power; tin economy
Fibre splice loss	Fusion ~0.01–0.1 dB < mechanical ~0.1–0.5 dB < connector ~0.2–0.5 dB
Fibre/joint cleaning	≥ 99 % isopropyl alcohol + lint-free wipes; D-Gel for jelly-filled cable

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End of Chapter 1. Next: Chapter 2 — Electronic Devices & Circuits.